

MPFM — Information-Projective Framework (V2)

A Quantitative Extension with Spectral Dimension and Neutrino Mass Prediction

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Abstract

We present Version 2 (V2) of the Information-Projective Framework (MPFM), a minimal extension of the previously established axiomatic model. This version introduces a concrete spectral operator acting on an informational space with non-integer spectral dimension $D_s < 3$. Without introducing free parameters, V2 yields a quantitative prediction for the sum of neutrino masses, $\Sigma m_\nu \approx 0.06$ eV, directly linked to the cosmological constant Λ . The framework remains fully compatible with known physical laws while providing a falsifiable, experimentally testable prediction.

1. Introduction

Modern physics successfully describes a wide range of phenomena, yet foundational questions remain unresolved. The nature of time, the origin of irreversibility, the hierarchy of fermion masses, and the role of information are still treated as emergent or auxiliary concepts. The MPFM framework reverses this logic by positing information as fundamental and dynamics as emergent.

2. Informational Space and Projection

The framework postulates an abstract informational space F , from which observable spacetime emerges via non-faithful projection. This projection necessarily induces information loss, providing a natural origin for irreversibility, decoherence, and noise.

3. Spectral Dimension and Fractality

A key ingredient of V2 is the introduction of a non-integer spectral dimension D_s . Empirical and theoretical considerations motivate a value $D_s = 2.90$, corresponding to a small but finite deficit relative to three-dimensional space.

4. Informational Spectral Operator

We define a minimal informational spectral operator Δ_I acting on a reduced informational Hilbert space. Its spectrum reflects the fractal nature of information under projection and does not rely on spacetime geometry.

5. Fixing the Informational Scale

The informational length scale ℓ_I is uniquely identified with the inverse square root of the cosmological constant, $\ell_I = \Lambda^{-1/2}$, introducing no additional free parameters.

6. Neutrinos as Near-Preprojective Fermions

Neutrinos occupy a unique position within the framework due to their weak coupling and near absence of charge. They remain closest to the informational space, making them ideal probes of the underlying structure.

7. Quantitative Prediction of Neutrino Mass Sum

Using the spectral properties of Δ_I , the framework predicts a fundamental mass scale proportional to $(3 - D_s)/\ell_I$. This yields a robust prediction for the sum of neutrino masses: $\Sigma m_\nu \approx 0.06$ eV.

8. Comparison with Observational Constraints

The predicted value lies below current Planck bounds and within the sensitivity range of forthcoming surveys such as Euclid and SKA, rendering the framework directly testable.

9. Relation to Mixing and CP Violation

Large PMNS mixing angles and leptonic CP violation arise naturally from maximal projective misalignment.

10. Scope and Limitations

While V2 provides a quantitative prediction for neutrino masses, other fermion masses remain qualitatively explained but not yet calculable.

11. Falsifiability and Experimental Outlook

Future cosmological measurements will decisively test the predicted neutrino mass sum. A significant deviation would falsify V2 without ambiguity.

12. Conclusion

MPFM V2 elevates the framework from a purely explanatory model to a quantitatively predictive theory. Its success or failure will be determined by forthcoming data.